

DATA70302 Assignment 2 2024/25

In this assignment you are asked to conduct a Topological Data Analysis (TDA) of Google Trends data. You should choose one of the topics listed and identify appropriate keywords within the topic. The topics are detailed below. Your analysis must contain the key elements which are explained below. The deadline for this assignment is **Friday 2nd May 2025 at 15:00**.

Data

This assignment requires the use of `pytrends` and `gudhi` libraries in Python. Data for Google Trends should be obtained using `pytrends`. Each keyword will produce a weekly time series of interest, where interest ranges from 0 to 100. The `related_topics()` functionality will allow the identification of related topics for analysis. Please see Computer Laboratory 8 for the process of downloading Google Trends Data.

The topics from which you should select are:

- Human Rights (Dancy and Fariss, 2024). A suggested set of search terms are provided within the article.
- Political issues and elections (Mellon, 2013). This is an early study of Google Trends, but with recent elections in the UK and US, an updated analysis can be undertaken on similar lines.
- New ways of working and burnout (Aguilera et al., 2021). Follows the argument that we search for things that will help us improve our lives and deal with challenges like burnout. This paper uses regional series.
- Social determinants of health (Di Simone et al., 2024). Looks at searches for terms that link to poor health. The article also produces multi-dimensional point clouds that link to our work in earlier lectures.

Each topic has a relevant reference listed to aid your search. When downloading data you should focus on either weekly, or daily frequency. Use of weekly frequency observations for the most recent 5 years will give a total of 260 observations for the analysis. If using daily frequency then you should consider reducing the time frame to avoid your analysis becoming overly complex computationally.

Empirical Elements

In order to complete this assignment you will need to calculate the following elements:

1. Time series for keyword and 5 related words (either `related_queries` or user input)
2. Point cloud for keyword and lag for keyword (Cloud A)

3. Point cloud for keyword and **top related word** (Cloud C)
4. **Time series of persistence norms** for 52 week window with:
 - A: Keyword plus 1 lag
 - B: Keyword plus 4 lags
 - C: Keyword plus 1 related search
 - D: Keyword plus 5 related searches
5. **Time series of Wasserstein/Bottleneck distances** between persistence diagrams with 1-week increments for A to D

These elements contribute to the Data and Results sections of the assignment. Inclusion of all of the elements is required to obtain full marks in the two empirical sections. Please note that the individual elements do not carry a specific number of marks. The mark you obtain will depend on the understanding shown and the evaluation of the outcome of each element. Full mark descriptors are in the next section.

Submission Elements

The elements which should appear in the analysis are detailed in Table 1. You may choose to adopt a different structure and/or use the word counts in a different fashion. The only formal requirement is that the **total number of words cannot exceed 1000.**

Table 1: Elements of Report

Section	Words	Contents
Introduction	100	Set out the aim of the study, the topic and the motivation for using Topological Data Analysis
Literature Review	250	Focus on existing studies with TDA. Short discussion of relevant application to keyword
Methodology	150	TDA, norm construction and Wasserstein/Bottleneck Distance (if used)
Data	200	Exploratory data analysis, plots of point clouds, correlation matrices for larger clouds, summary statistics table
Results	200	Time series of signals and interpretation
Conclusion	100	Relationship of results to literature, critical evaluation of the inference from your results, next steps

Further, you are asked to **include an appendix which contains your code for the assignment.** The appendix does not count towards the final mark, and is not included in the word count.

Marking

This assignment is worth **40%** of the grade for DATA70302. The total mark you receive will be out of 100. The weighting will be applied in the calculation of your final grade. Grading will follow the standard grade descriptors. No marks are assigned to individual sections of the report.

Grade Descriptors

The text in this section is taken from the MSc Data Science handbook and is included for students on other programmes

Below is a set of general criteria that will guide the assessments on the program. As the assessments are quite diverse in form it is not possible to be precise about these criteria at this level. So, you should understand these as expressing the spirit of the criteria. The assessment requirements for each course unit are specified in the individual course guides: Please make sure that you carefully read the assessment-related part of your course unit guide.

High Distinction (80+)

This is outstanding work in every respect. It reaches all the standards of Distinction work and, additionally, reveals extensive knowledge of the topic and demonstrates originality. It is excellent in both rigour and organisation. It is very well-written. It demonstrates clear potential for the student to make an original contribution to knowledge at PhD level.

Distinction (70-79)

This is excellent work, showing evidence of comprehensiveness and appropriate focus. Where relevant to the assessment type, it covers a wide range of the relevant literature, which is synthesized into a high-quality analysis. It will be methodologically and/or theoretically sophisticated. It endeavours to develop an original approach to assignment topic. There are few weaknesses.

Merit (60-69)

Work in this range is clearly competent and presented well, with work in the upper end of the range demonstrating the capacity to undertake further postgraduate research. Analyses are soundly conducted. The standard of writing is good. There may be occasional weaknesses in coherence of text and occasional errors in analysis particularly at the bottom end of the mark range. The choice of method to tackle the problem will be fit for purpose although perhaps not showing the sophistication expected for a distinction.

Pass (50-59)

This represents the minimum performance required on a Masters course. Structure and presentation should be satisfactory, demonstrate an awareness of literature (where relevant

to the type of assessment). Analyses have been conducted mostly without error, though non-optimal methods may have been chosen.

Fail (40-49) (compensatable)

Does not reach the level required for a Masters course. It discusses relevant material, although it is descriptive and lacks analytical depth. There are some errors of fact and/or inconsistencies in the analysis. It may be reasonably presented and demonstrate an awareness of data science methods and techniques but fails to effectively address the assignment topic.

Fail (0-39)

Work in this range does not reach MSc standard. The work does not constitute a sufficient answer to the question set or assignment topic. It is poorly organised and exhibits numerous errors and major inconsistencies.

Submission Information

The text in this section is taken from the MSc Data Science handbook and is included for students on other programmes

The following general summary applies:

- All assignments must be submitted to Turnitin via Blackboard by the deadline stated.
- Problems - If you are encountering any problems, please see either your course unit convener, pathway lead or the Programme Director.
- Length of written coursework - Course unit conveners will state the specific length limits for individual pieces of work. Except for your extended research project report, word counts may exceed the limit by 10% of the prescribed word limit without penalty. Work that exceeds this will be penalised. Word limits do not include bibliographies (or appendices if used) but do include footnotes and endnotes.
- Bibliography & Referencing - a full bibliography should be appended listing all sources consulted in preparing the assessment assignment. We do not stipulate you must use one form of referencing or another, so long as you are consistent and thorough in providing all the relevant publication details and at the necessary locations in the text. Choose one referencing and bibliography style and follow it consistently.

Coursework Submission Details

The University now uses electronic submission only for all assessed work and the deadline is 3pm unless otherwise stated.

When submitting to Turnitin for the 'Submission Title' please ONLY enter your 7 or 8 digit ID number. DO NOT enter anything else in this box. If we are unable to identify your

work by ID number, you may be counted as not having submitted. You should also consult your course Blackboard site for the submission deadline dates and submission details.

How to submit your work:

1. Log into Blackboard via My Manchester at: <https://login.manchester.ac.uk> and go to the course unit site.
2. Click on the 'Assessment' folder from the menu on the left hand side.
3. You will then see a folder named 'Coursework Submission'. Click on this and you will be taken to the coursework folder which contains full instructions for online submission.
4. Click on 'Submit your DATA70302 Assignment 2 here'

Before Submitting Remember:

- When submitting online please ensure you submit the correct version of your work.
- Written coursework must be double-spaced and 12-point type.
- Ensure pages are numbered and that your University ID number (seven/eight-digit library card number) and the relevant course code (DATA70302) appear on each page.
- The total number of words (excluding the bibliography/final list of references and any appendices but including any footnotes) must be printed at the end of the essay/report. Note that even though your name is automatically entered when submitting your coursework (step 1 of 2), assignments are marked anonymously so your name will not be visible to the marker.

For further information please see "A Student Guide to Submitting an Assignment via Turnitin": <http://documents.manchester.ac.uk/display.aspx?DocID=13010>

Word limits:

- For all written assessed work, it is important to comply with the word limit given in the course guide. You may be penalised up to 5 points if your assessed work is significantly longer than specified. As a guide, an acceptable variation is 10 percent above the word limit.
- You must indicate the word count at the end of your essay/report. Failure to do so will result in a deduction of 2 marks.
- Your bibliography and appendices (if used) these are not counted in the final word count.

Late Submission:

- Late Submission of Coursework will be penalized by 10 marks per day penalty. More information on the Policy on Submission of Work for Summative Assessment on Taught Programmes can be found on the Teaching and Learning Support Office website at: <http://www.tlso.manchester.ac.uk/map/teachinglearningassessment/assessment/sectionb-thepracticeofassessment/policyonsubmissionofworkforsummativeassessment/>

- Late submission penalties will be applied by the programme administrator before your marks are uploaded on campus solutions. Marks that appear on blackboard/Turnitin are the marks before any such penalties are applied.

Assessment for Students with Disabilities:

- The University has responsibilities under the Disability Discrimination Act to make reasonable adjustments to its provision, including methods of assessment, to ensure that students with disabilities are not disadvantaged for reasons relating to their disability.
- Further information is available in the Guidance on Assessment for Students with Disabilities.

Frequently Asked Questions

- **What if there are issues with downloading data?:** This assignment requires the use of `pytrends` for the downloading of Google Trends data. In the event that you are unable to get data, the most common issue is a time out. This can be resolved within minutes, or by trying again on another computer/day. You may also use packages in R to download Google Trends data then perform the analysis in Python.
- **Can alternative data frequencies be used?:** You may use alternative frequencies of data, but you are responsible for ensuring that the correct data is downloaded

References

- Aguilera, A. M., Fortuna, F., Escabias, M., and Di Battista, T. (2021). Assessing social interest in burnout using google trends data. *Social Indicators Research*, 156:587–599.
- Dancy, G. and Fariss, C. J. (2024). The global resonance of human rights: What google trends can tell us. *American Political Science Review*, 118(1):252–273.
- Di Simone, E., Panattoni, N., Dionisi, S., Giannetta, N., Renzi, E., Massimi, A., Di Muzio, M., and Fabbian, F. (2024). Social determinants of health and google trends: a worldwide infodemiological report. *European Review for Medical & Pharmacological Sciences*, 28(10).
- Mellon, J. (2013). Where and when can we use google trends to measure issue salience? *PS: Political Science & Politics*, 46(2):280–290.